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Sutra Mono

A monospace drawn as a single thread. One centre-line, one pen, nothing that is there for show.

3 weights · 358 glyphs · 600/1000 em · truetype

designed by naren naik

prologue

The brief was *one sentence long.*

“Smooth, with a flow between the characters. Nothing unessential. Rounded, and feeling complete, like San Francisco.”

Most coding fonts begin from a sans-serif and squeeze it into a fixed grid. Sutra began from a different question: what is the least you could draw and still have every letter read? The answer turned out to be a single line per letter, the path a pen would take if it never lifted, and a single nib to trace it.

The name comes from Sanskrit. *Sūtra* means thread, and by extension a short rule, the compressed aphorisms that hold a whole system in a line. Code is that kind of writing. A function signature, a type, a composition such as `cata = alg ∘ fmap cata` carries an argument in a dozen characters. The font is meant to stay out of the way of that.

This document follows the decisions in the order they were made: the pen, the curve, the grid, the letters that share their parts, the symbols that make it a font for people who compose functions, and finally what the first version taught us about the gap between geometry and the eye.



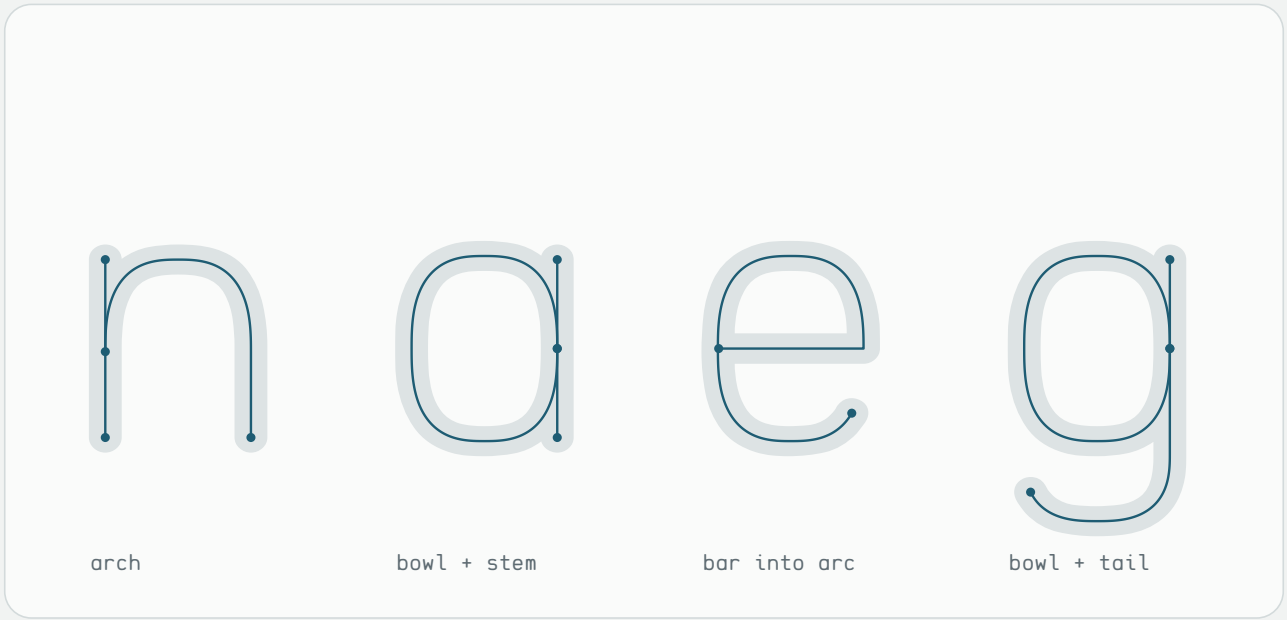
The word “sutra” as its skeletons: the thread before the pen.

chapter 1

One line, one pen

Every glyph in Sutra is stored as a *skeleton*: a few centre-line paths, sometimes a dot. Nothing about weight lives in the drawing. The outline you see is produced by sweeping one pen along each path, with round caps where a stroke ends and round joins where it turns.

Mathematically it is a Minkowski sum, the set of every point within one pen-radius of the line. In practice it means three promises are kept automatically. Every terminal ends in the same soft half-circle. No corner is ever sharp. And weight is a single number.



Skeletons in blue over the stroked result. The pen radius does the rest.

weight = pen radius

cata ∘ fmap — light 58u

cata ∘ fmap — regular 82u

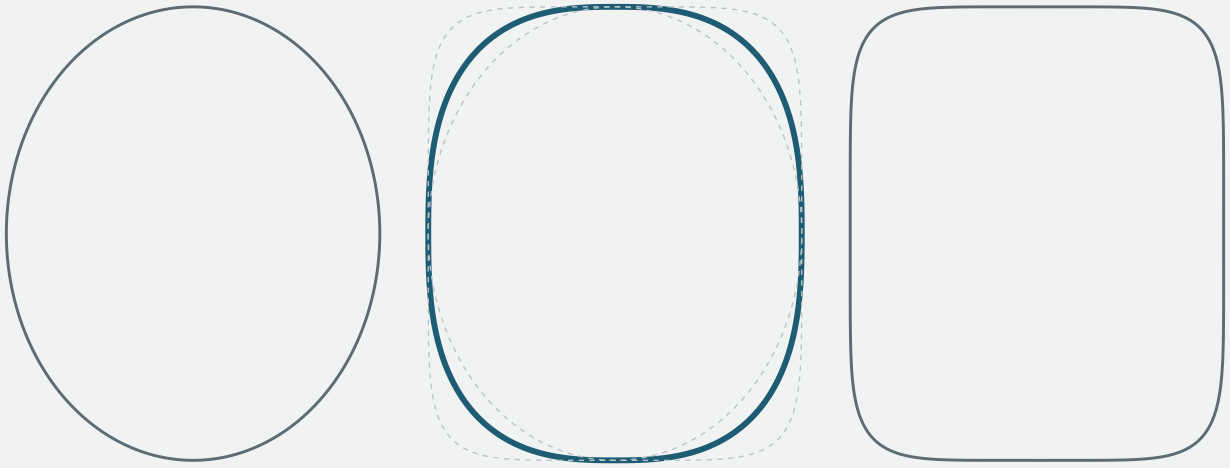
cata ∘ fmap — bold 114u

chapter 2

The soft square

San Francisco feels calm partly because its rounds are not circles. They are flatter at the sides and fuller at the shoulders, somewhere between a circle and a box. Sutra makes that shape explicit with a superellipse, a curve the Danish poet and designer Piet Hein made famous for a Stockholm square.

$$|x|^n + |y|^n = 1$$



n = 2 circle

n = 2.7 sutra

n = 6 box

At n = 2 the curve is a circle; as n grows it approaches a box. Sutra uses 2.7: straight enough that the sides of o read as walls of a column of text, round enough that nothing has a corner. JetBrains Mono reaches a similar place by eye; it argues that near-rectangular ovals make the text pattern clearer and leave open lanes for the eye moving down a file.

One curve serves ten letters and four figures: o b d p q a g, 0 C G 0 6 8 9. Round forms also rise about 1.7 percent above and below the flat letters, an overshoot, so that an o beside an x looks the same height rather than smaller.



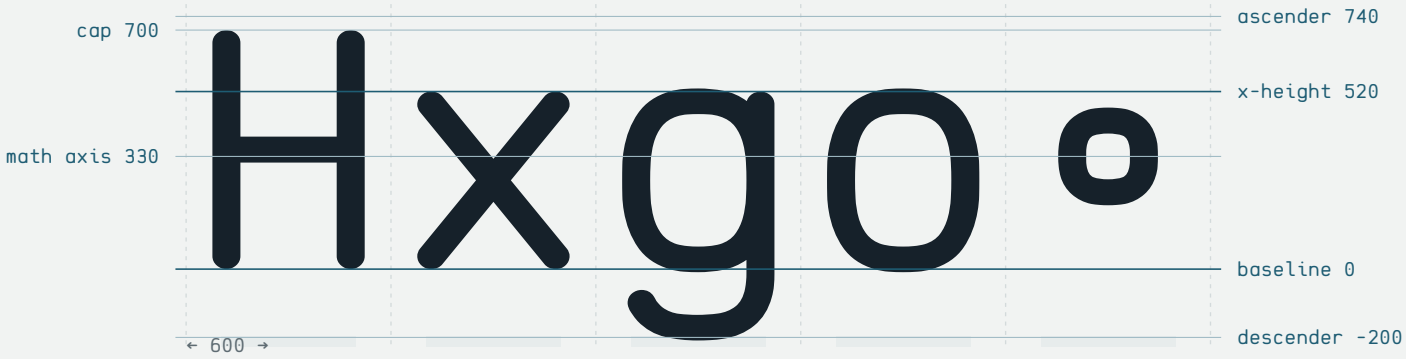
The o crosses both lines by nine units. The x does not.

chapter 3

Six hundred units

A monospace font makes one promise above all others: every character takes the same width, so a column of code stays a column. Sutra's cell is 600 units of a 1000-unit em. Inside that cell the stems sit at 118 and 482, which leaves the same breathing room on either side of every letter.

The x-height is set high, at 520 units, so lowercase letters fill more pixels at small sizes without making lines longer. Capitals stand at 700, ascenders at 740, descenders drop to −200. The math axis, where `+ - = ∘ →` sit, is at 330, the optical middle of the lowercase rather than its geometric centre.



Dotted lines mark the fixed cells; the grey bars under each cell show the stem positions, 118 to 482.

```
let advance = 600   -- every glyph
let xh      = 520   -- lowercase
let capH    = 700   -- capitals
let axis    = 330   -- operators
let pen     = 82    -- regular
```

Because every character is 600 units wide, the equals signs and comments line up without any help from the editor.

chapter 4

One bowl, seven letters

Once the curve existed, the lowercase almost drew itself. The same squircle bowl, with a straight stem placed on its left or right, rising or descending, gives seven letters. The `a` and `g` are single-storey for the same reason: a two-storey `a` would need a second, smaller bowl and a hook, which is two more ideas than the letter needs.

This is the font's version of "nothing unessential". A reader learns one shape and gets seven letters back, and the letters agree with one another because they are, literally, the same object.



Blue: the shared bowl. Ink: the stem or tail that turns it into a letter.

The same economy runs through the rest of the alphabet. `n`, `h` and `m` share one arch; `u` is that arch turned over. `i` and `l` end in a small tail that curls into the next character, which is where the "flow between letters" in the brief lives: it fills the empty space a narrow letter leaves in a wide cell and leads the eye forward.

nhmu il

chapter 5

Told apart at a glance

In prose, context forgives a lot: nobody misreads “l00k” because they know the word. In code there is no such mercy. `O` and `0`, `l` and `1`, a comma and a period, each confusion is a bug. Sutra resolves each one with the smallest difference that works.



Zero is narrower than O and carries a single dot. No slash, so it never looks like Ø.



Capital I has bars. Lowercase l ends in a tail. One has a flag and a foot. The pipe runs the full height.



The r shoulder rolls over and turns down early, so rn never fuses into m.

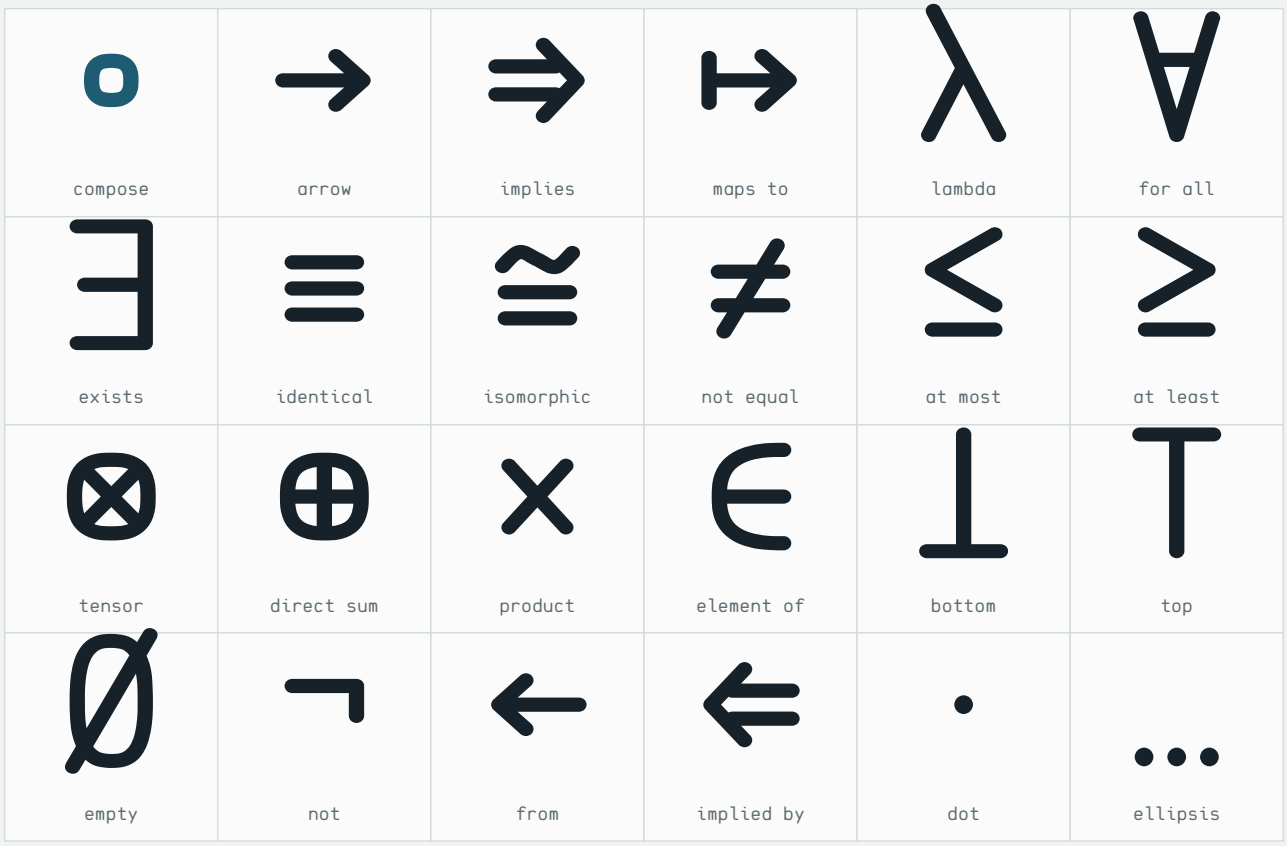


Dots are a touch heavier than the stroke, so punctuation survives at eleven pixels.

chapter 6

The composition set

Most coding fonts treat mathematical symbols as an afterthought, borrowed from a fallback font in a different weight and sitting at a different height. Sutra draws them with the same pen, on the same axis, as first-class citizens. If you write `g ∘ f`, the ring sits exactly where the eye expects an operator, and it has the same stroke as the letters on either side.




```
cata alg = alg ∘ fmap (cata alg) ∘ unfix
∀ f g h. (f ∘ g) ∘ h ≡ f ∘ (g ∘ h)
```

A catamorphism and the associativity law, set in Sutra Mono Regular.

chapter 7

Where it lives

A coding font is read for hours at a time, mostly at thirteen or fourteen points, mostly on a dark background. This is the page that matters. Read it at arm’s length and notice what you do not notice: no terminal catches the eye, no letter is louder than its neighbours.

pipeline/compile.ts

```
-- ad format spec ⇒ deterministic Moore machine
compile = emit ∘ minimise ∘ determinise ∘ parse

fold :: (F a → a) → Fix F → a
fold alg = alg ∘ fmap (fold alg) ∘ unfix

const step = (s: State, e: Event): State =>
  s.on[e.kind]?.(e) ?? s;   // total, never throws

if (bid >= floor && fill != 0) {
  render("banner");
}

// 00 o Il1| {}[]() #$%&@ → ← ↦ ⊗ ⊕ ⊥ ⊤ ∈ ≅ ¬
```

Sutra Mono Regular at 12.6 pt on a night background, with a quiet syntax palette.

chapter 8

What the eye wants

Version one was honest geometry, and the first round of use showed exactly where geometry and perception part ways. Type designers have known these things for a long time; Sutra learned them by drawing, and version 1.100 applies each one.

01 Horizontals look heavier than verticals.

A round pen gives every direction the same width, so bars and the tops of bowls read slightly heavy. Sutra keeps the single-pen idea but changes the nib: an ellipse 8 percent wider than it is tall, which thins horizontals and sets diagonals between the two automatically.

02 Curves need matching curvature, not only direction.

Outlines are now fitted to clean Bézier curves with a node at every extreme, and the spine of the s is solved so that it meets both bowls tangentially, without a kink.

03 An s is not two stacked c’s.

The first one felt as if it leaned left. The redrawn s has a slightly larger lower half and a straight diagonal spine, so the two halves look equal and the letter stands upright.

04 Narrow letters need a job in a wide cell.

The first r looked like an n that had lost a leg. It now has a rounder shoulder that finishes with a small downturned terminal, filling its cell without borrowing anyone else’s.

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next

- Texture healing: contextual alternates that let i and l lend space to m and w, without breaking the grid.
- Optional functional ligatures for -> => != >=, nothing decorative.
- A variable weight axis, with 400 recommended on dark backgrounds and 450 on light.
- Submitted to Google Fonts under the SIL Open Font License (google/fonts #11111).

colophon

Specification

family	Sutra Mono
weights	Light (pen 58) · Regular (82) · Bold (114)
glyphs	358 · GF Latin Core plus the composition set
em / advance	1000 / 600
vertical metrics	x 520 · cap 700 · asc 740 · desc −200 · axis 330
curve	superellipse, n = 2.7, overshoot 9 units
format	TrueType, fixed pitch
source	github.com/narennaik/sutra-mono

install

Install the three .ttf files, then set `"editor.fontFamily": "Sutra Mono"` in VS Code, or `font-family = Sutra Mono` in Ghostty. Recommended size 13 to 15 pt with a line height of about 1.5.

character set

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789 .,:;!?'"`
()[]{}<>/\|_^~*+-=#$%&@
∘→←⇒⇐↦λ∀∃≡≠≅≤≥⊗⊕×∈⊥⊤∅¬·…

Set in Sutra Mono and Newsreader. Glyphs in this document are drawn directly from the shipped font outlines; construction diagrams are drawn from the same skeletons that generated them.